

Advanced (Carolina Reaper)

Q1. What is the primary purpose of a box-and-whisker plot?

- (A) To display the mean of a dataset
- (B) To show the median and quartiles of a dataset
- (C) To compare the range of two datasets
- (D) To identify the mode of a dataset
- (E) To calculate the standard deviation of a dataset

Q2. Which part of a box-and-whisker plot represents the interquartile range (IQR)?

- (A) The box
- (B) The whiskers
- (C) The median line
- (D) The outliers
- (E) The scale

Q3. A dataset has $Q_1 = 12$ and $Q_3 = 20$. What is the IQR ?

- (A) 4
- (B) 8
- (C) 12
- (D) 16
- (E) 8

Q4. What is the formula to calculate the IQR ?

- (A) $IQR = Q_3 - Q_1$
- (B) $IQR = Q_1 - Q_3$
- (C) $IQR =$

$\frac{Q_3 + Q_1}{2}$

(D) $IQR =$

$\frac{Q_3 - Q_1}{2}$

(E) $IQR = Q_3 \cdot Q_1$

Q5. How do you determine if a data point is an outlier using the $1.5 \cdot IQR$ rule?

- (A) If the data point is within $1.5 \cdot IQR$ of Q_1 or Q_3 , it

is an outlier

(B) If the data point is beyond $1.5 \cdot IQR$ from Q_1 or Q_3 , it is an outlier

(C) If the data point is exactly $1.5 \cdot IQR$ from Q_1 or Q_3 , it is an outlier

(D) If the data point is within $1.5 \cdot IQR$ of the median, it is an outlier

(E) If the data point is beyond $1.5 \cdot IQR$ from the median, it is not an outlier

Q6. A box-and-whisker plot shows $Q_1 = 10$, $Q_3 = 18$, and an outlier at 25. What is the value of $1.5 \cdot IQR$?

- (A) 6
- (B) 8
- (C) 10
- (D) 12
- (E) 14

Q7. Which of the following box-and-whisker plots indicates the largest dispersion?

- (A) One with a short box and short whiskers
- (B) One with a long box and short whiskers
- (C) One with a short box and long whiskers
- (D) One with a long box and long whiskers
- (E) One with no whiskers

Q8. What does a box-and-whisker plot with no outliers indicate?

- (A) The data is normally distributed
- (B) The data is skewed to the left
- (C) The data is skewed to the right
- (D) The data has a large range
- (E) The data has a small range

Open Ended Questions (FRQ)

Q9. Construct a box-and-whisker plot for the dataset: 12, 15, 18, 20, 22, 25, 30, 35, 40. Identify Q_1 , Q_3 , and the IQR . Check for outliers using the $1.5 \cdot IQR$ rule.

Q10. Compare the dispersion of two datasets using box-and-whisker plots. Dataset A: 10, 12, 15, 18, 20. Dataset B: 8, 10, 12, 15, 18, 20, 25. Identify which dataset has the larger dispersion and explain why.

Chef's Tips

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- Tip 1: Emphasize the importance of accurately calculating quartiles and the IQR .
- Tip 2: Stress the role of box-and-whisker plots in visualizing dataset dispersion and identifying outliers.
- Tip 3: Encourage students to consider the implications of outliers on dataset analysis.